Displacement of Vancouver Island: paleomagnetic evidence from the Karmutsen Formation[1]

R. W. YOLE

Department of Geology, Carleton University, Ottawa, Ont., Canada K1S 5B6

AND

E. IRVING

Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa, Ont., Canada K1A 0E4

Received December 17, 1979

Revision accepted May 6, 1980

New paleomagnetic results from the Karmutsen Formation (Late Triassic) of Vancouver Island confirm the presence of two families of magnetizations (*X* and *Y*), both of which are inconsistent with known Mesozoic and Cenozoic geomagnetic fields of cratonic North America. The *X* magnetizations have coherent directions with the exception of a subset of five sites (the B subset). We argue that the deviation of the B subset is caused either by a 31 ± 13° anticlockwise rotation of a small block relative to the main sampling areas or by a short-term excursion of the field. The *X* magnetization has an overall mean direction 008°, −33° $\alpha_{95} = 6°$ based on results from 147 oriented samples (usually 2 specimens from each) collected at 28 sites spanning about 6000 m stratigraphically. We interpret this as the original Late Triassic magnetization. The corresponding *X* paleopole (21°N, 44°E $A_{95} = 6°$) is strongly far-sided and right-handed with respect to the Mesozoic apparent polar wander path for cratonic North America. The paleolatitude indicated for Vancouver Island in the Late Triassic is either 18°N or 18°S, the latter being preferred on the grounds that it yields a more consistent pattern for Cordilleran magnetizations, but the ambiguity is still not settled. In either case the results show that Vancouver Island was far south of its present position relative to North America in the Late Triassic, thus confirming the previous results of Irving and Yole. The *Y* magnetizations, with more heterogenous properties, occur at 14 sites (66 oriented cores, usually 2 specimens each). *Y* magnetizations are generally softer than *X* and for this and other reasons we regard them as secondary and post-Triassic in age. Individual site poles for the *Y* magnetization are, with minor exceptions, right-handed and slightly far-sided with respect to the apparent polar wandering path for cratonic North America. The mean paleopole for *Y* magnetizations is situated at 70°N,15°W $A_{95} = 11°$. Both the *X* and *Y* magnetizations are consistent with either northward motion of the westernmost Cordilleran elements accompanied by clockwise rotation, or with oblique translation from the southwest. The northward component of motion derived from *X* directions would be the same in both instances and amounts to 1300 or 4900 km depending on whether the northern or the southern paleolatitude option is chosen. Our preference is for the latter and we present arguments which suggest that Vancouver Island may have been originally derived from a region near to eastern Gondwana or from a block east of Gondwana that might have included Malaysia. The procedures used for the tectonic analysis of aberrant paleopoles are described in the Appendix.

De nouveaux résultats paléomagnétiques provenant de la formation de Karmutsen (fin du Trias) dans l'île de Vancouver confirment la présence de deux familles d'aimantations (*X* et *Y*), et les deux ne correspondent pas aux champs géomagnétiques connus pour le Mésozoïque et le Cénozoïque du craton de l'Amérique du Nord. Les aimantations *X* ont des directions cohérentes à l'exception d'un sous-ensemble de cinq sites (le sous-ensemble B). Nous proposons ici que la déviation du sous-ensemble B est causée soit par une rotation antihoraire de 31 ± 13° d'un petit bloc par rapport aux principales zones d'échantillonnage, soit par une excursion de courte durée du champ. L'aimantation *X* a une direction moyenne globale de 008°, −33° $\alpha_{95} = 6°$, basée sur les résultats de 147 échantillons orientés (habituellement deux spécimens pour chaque échantillon) provenant de 28 sites recouvrant une épaisseur stratigraphique d'environ 6000 m. Nous interprétions cela comme l'aimantation originelle de la fin du Trias. Le paléopôle *X* correspondant (21°N, 44°E, $A_{95} = 6°$) est fortement déplacé et à droite par rapport au trajet apparent de migration polaire pour le craton de l'Amérique du Nord au Mésozoïque. La paléolatitude pour l'île de Vancouver à la fin du Trias serait soit 18°N, soit 18°S, probablement cette dernière en se basant sur le fait qu'elle donne une image plus compatible avec les aimantations de la Cordillère, mais l'ambiguïté n'est pas encore résolue. Dans l'un ou l'autre cas, les résultats montrent que l'île de Vancouver était beaucoup au sud de sa position actuelle par rapport à l'Amérique du Nord à la fin

Can. J. Earth Sci., **17**, 1210–1228 (1980)

[1]Earth Physics Branch Contribution No. 871.

0008-4077/80/091210-19$01.00/0

©1980 National Research Council of Canada/Conseil national de recherches du Canada

du Trias, ce qui confirme les résultats antérieurs de Irving et Yole. Les aimantations *Y*, avec des propriétés plus hétérogènes, se rencontrent en 14 localités (66 carottes orientées, deux spécimens pour chacune). Les aimantations *Y* sont généralement plus faibles que les aimantations *X* et pour cela et d'autres raisons, nous les considérons comme secondaires et d'âge post-Trias. Les sites de pôles individuels pour les aimantations *Y* sont, avec des exceptions mineures, à droite et un peu en dehors par rapport au trajet apparent de migration polaire pour le craton de l'Amérique du Nord. Le paléopôle moyen pour les aimantations *Y* se situe à 70°N, 15°O, $A_{95} = 11°$. Les aimantations *X* et *Y* sont toutes deux compatibles soit avec un mouvement vers le nord des éléments de l'extrême ouest de la Cordillère, accompagné d'une rotation horaire, soit avec une translation oblique à partir du sud-ouest. La composante nord du mouvement déduite des directions *X* serait la même dans les deux cas et serait de 1300 ou de 4900 km selon qu'on choisit la position nord ou sud de paléolatitude. Notre préférence va à la seconde option et nous présentons des arguments qui suggèrent que l'île de Vancouver a pu dévier originellement d'une région située près de l'est de Gondwana ou d'un bloc à l'est de Gondwana qui a pu inclure la Malaysie. On décrit en appendice les procédures utilisées pour l'analyse tectonique des paléopôles aberrants.
[Traduit par le journal]

Introduction

Some years ago, we presented paleomagnetic results from the Karmutsen Formation and suggested that, in the Late Triassic, Vancouver Island was situated several thousands of kilometres south of its present position relative to North America (Irving and Yole 1972). Results from 11 sites in the Buttle Lake area (Fig. 1) showed that Karmutsen rocks contain a magnetization (here called *X* but referred to in our original paper as *B*) with directions grossly incompatible with the expected geomagnetic field for this region in the Late Triassic. The *X* magnetizations are consistent in direction through a great stratigraphic thickness and are characterized by high unblocking temperatures and fields (coercivities). The unblocking temperatures and fields are a measure of the "hardness" of the remanent magnetization (Zijderveld 1975). We argued that *X* was the original magnetization acquired at the time the rocks were formed. We also showed that other magnetizations (here called *Y* but referred to previously as *A*) with less consistent directions and lower unblocking temperatures and fields were present. We argued that *Y* was a secondary overprint, probably acquired during the Jurassic, because it was less hard and because the directions were roughly along the expected Jurassic field.

Symons (1971*b*) has also studied the Karmutsen (15 sites from 4 areas). He found both *X* and *Y* magnetizations, but stated that they had "equal stability." He listed several possible interpretations, including the displacement of Vancouver Island, but found it impossible to decide which was correct. There would, at first sight, seem to be a conflict between our results and those of Symons, but this is not so. Symons' discussion of stability focussed on the consistency or otherwise of the magnetization directions during demagnetization in alternating fields (AF). As our original results also showed both *X* and *Y* are directionally stable, although *X* is more internally consistent than *Y*, as Symons also found. The important difference between *X* and *Y* disclosed by our work was in their magnetic properties; *X*, with its high unblocking temperatures and coercivities indicative of thermoremanent magnetization (TRM) or high-temperature chemical remanent magnetization (CRM) is, we argued, likely to be the original magnetization, whereas *Y*, with its low coercivities and unblocking temperatures indicative of viscous remanent magnetization (VRM) or viscous partial TRM is more likely to be secondary. Polished sections provided key evidence. The *X* magnetization is associated with iron minerals containing fine-grained high-temperature exsolution intergrowths of ilmenite and magnetite, whereas *Y* is associated with nonlamellar titanomagnetites (Irving and Yole 1972). The two magnetizations observed, with their variable unblocking fields and temperatures, are thus a reflection of the variable oxidation state of the Karmutsen Formation. These were the reasons that we originally gave for arguing that the *X* magnetization, found in rocks whose magnetic minerals had undergone high-temperature deuteric alteration, was the original magnetization, thus requiring that Vancouver Island had been displaced about 4500 km from the south relative to North America since the Triassic (Irving and Yole 1972).

Stratigraphic equivalents of the Karmutsen occur in the Queen Charlotte Islands and in the Wrangell Mountains of southern Alaska. In the Queen Charlotte Islands, the Karmutsen is metamorphosed to greenschist or, in some places, amphibolite facies, and the magnetizations are apparently of Tertiary age (Hicken and Irving 1977). In the Wrangell Mountains, the Nikolai Greenstone has been studied by Hillhouse (1977). The Nikolai consists largely of terrestrial basalt flows in which *X* type properties are present. The paleomagnetic

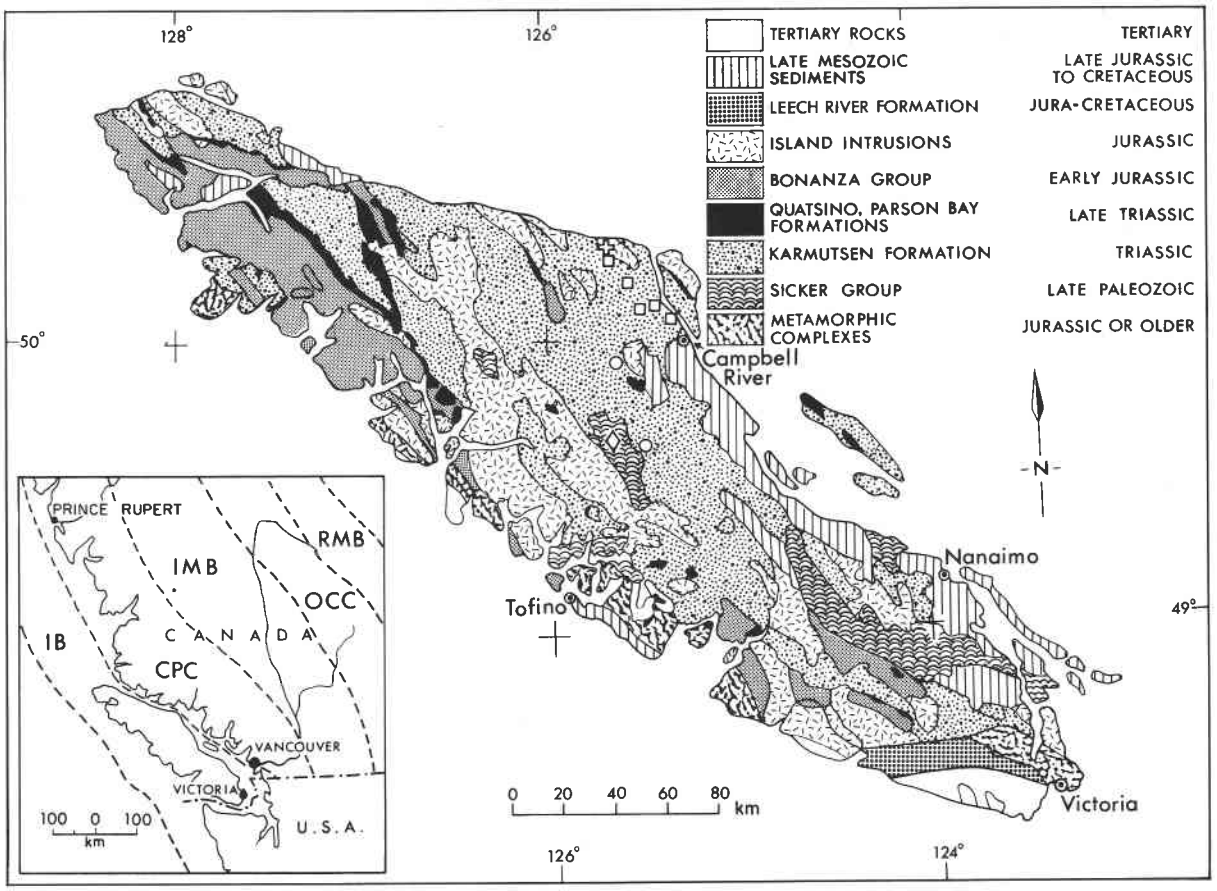


FIG. 1. Index map and generalized geology of Vancouver Island, after Muller (1977*a*). Sampling areas: diamond, Marble Peak (M); circles, Buttle Lake (B); squares, Campbell River (C); crosses, Island Intrusion (G). IB, Insular Belt; CPC, Coast Plutonic Complex; IMB, Intermontane Belt; OCC, Omineca Crystalline Complex; RMB, Rocky Mountain Belt.

properties appear to be simpler than those of the Karmutsen. The paleolatitudes (Hillhouse 1977, p. 2588) are in accord with those we had earlier obtained from the *X* magnetization of the Karmutsen. The results from the Nikolai, which is the stratigraphic equivalent of the Karmutsen and is considered (Jones *et al.* 1977) to belong to the same tectonic belt or terrane (Wrangellia), provides strong evidence in favour of our initial contention that Vancouver Island has moved from the south. However, our first results from the Karmutsen were based on a limited section and few sites, and more extensive observations were required to verify the postulated displacement—a displacement that if correct is fundamental to our understanding of the tectonic history of the western Cordillera. The present paper is an attempt to fulfill this need. Some of our earlier sites have been resampled, and many new sites in different areas have now been included (a total of 28 sites from 3 areas) so as to span the stratigraphic thickness of the formation.

While this paper was under review the results of another Karmutsen study based on 13 sites were reported (Schwarz *et al.* 1980). These results will be considered together with our new data in later sections of this paper.

Geology

The Canadian Cordillera has been divided into five major longitudinal physiographic and tectonic belts (Fig. 1, inset; Douglas *et al.* 1970; Gabrielse and Wheeler 1972). The westernmost of these, the Insular Belt, includes Vancouver Island and the Queen Charlotte Islands and extends northward to southeastern Alaska. It is bounded on the east by the Coast Plutonic Complex. A thick assemblage of Paleozoic (Sicker Group) and early Mesozoic (Vancouver Group) sedimentary and volcanic rocks, intruded by Jurassic granitoid plutons (Island Intrusions), underlies most of the axial mountain ranges of Vancouver Island. Late Jurassic to Late Cretaceous clastic sediments flank this axial

uplift. A narrow belt of Tertiary sediments, the landward edge of a seaward-thickening prism, is sporadically exposed along the west coast. Tertiary basic volcanics and intrusions form the southern tip of the island (Fig. 1), and several small Tertiary intrusive bodies are found in the interior. Metamorphic complexes, possibly including rocks as old as Devonian (Muller 1977*b*, p. 2066), occur in the southeastern part of the island. There, they are associated with Paleozoic volcano-sedimentary sequences for which they may form "basement." Metamorphic complexes on the west coast are presumed to be migmatized Paleozoic and early Mesozoic rocks associated with Early Jurassic granitoid plutons (Muller 1977*b*, p. 2072).

A narrow belt of sheared or metamorphosed Jurassic and Cretaceous sediments with minor volcanics, exposed partly along the west coast near Tofino and partly in a fault-bounded block extending east–west across the southern part of the island (Leech River Formation), has been termed the "Pacific Rim Complex," and has been correlated with the Franciscan sequence of California. This forms the basis for a recent suggestion to split off this complex from the "Insular Belt" to form the inner part of a sixth major tectonic belt, the "Pacific Belt" of Muller (1977*b*, pp. 2072, 2076–2080, Figs. 6, 8). The outer part of Muller's Pacific Belt is represented on Vancouver Island by the earlier mentioned Tertiary basic volcanics and intrusives of the southern tip of the island.

Tectonic and stratigraphic analyses have led to "mobilistic" interpretations of the origin of the belts of western Cordillera, both in Canada and the United States. Paleozoic events have been suggested to include the opening of marginal basins behind magmatic arcs offshore from cratonic North America (Churkin 1974), or the eastward or northward drifting of arc complexes which became amalgamated in the Mesozoic with the craton (Monger *et al.* 1972; Jones *et al.* 1972). Large-scale post-Triassic movements, with strong northward components, have been indicated by paleomagnetic and other evidence (Tozer 1970; Monger and Ross 1971; Irving and Yole 1972; Burchfiel and Davis 1972; Monger 1977; Muller 1977*b*; Jones *et al.* 1977; Monger and Price 1979).

In parts of the Vancouver Island segment of the Insular Belt, Upper Paleozoic rocks of the Sicker Group are overlain by about 6000 m of basaltic pillow lavas, flows, pillow breccias, and minor sediments of the Karmutsen Formation. In some places, an unnamed unit of thinly-bedded clastic rocks, elsewhere found to bear Middle Triassic (Ladinian) fossils, intervenes between the Sicker Group and the Karmutsen in paraconformable succession (Yole 1965, 1969; Muller and Carson 1969; Carlisle 1972; Carlisle and Susuki 1974; Muller *et al.* 1974). The Sicker Group, the unnamed unit, and the lower part of the Karmutsen Formation are intruded by numerous sills of basalt or diabase, which have been considered to be co-magmatic with the Karmutsen lavas (Fyles 1955, pp. 10–11, 20; Carlisle 1972, p. 28; Muller 1977*a*, p. 42). As we show below, this geological conclusion is supported by the similarity in observed magnetization directions of sills intruded into Paleozoic rocks and of the Karmutsen lavas.

The age of the Karmutsen is well established from fossil evidence in the locally underlying unit (Late Ladinian), from Late Karnian fossils within intralava sediments (Dilleri Zone), and from the occurrence of Late Karnian fossils (Dilleri and Welleri Zones) in the conformably overlying Quatsino Formation (Carlisle and Susuki 1974, pp. 258–263; Jeletzky 1976, pp. 11–13). Carlisle and Susuki conclude that the Karmutsen lavas were extruded in the interval early Late Ladinian to early Late Karnian, possibly representing approximately 2.5–3.5 Ma (Carlisle and Susuki 1974, p. 268).

The ages of Island Intrusions (quartz diorite, granodiorite, and quartz monzonite) are also important in determining the sequence of events relating to the tectonic and magnetic history of Karmutsen rocks. K/Ar ages of the plutons range from 181–142 Ma (Early to Late Jurassic). According to Muller (1977*b*, pp. 2071–2072), the younger of these ages may not represent the times of intrusion of the plutons, most of which he believes were formed and emplaced in Early Jurassic time.

The present structural attitudes of Karmutsen rocks have been affected by several periods of deformation since the Late Triassic, but the most important is probably Late Mesozoic or Early Tertiary faulting and folding which has affected rocks as young as the Late Cretaceous in the sampling area (Sutherland-Brown 1966; Jeletzky 1976; Muller 1977*a*, *b*).

Sampling

The Karmutsen samples collected for this study are from three sampling areas denoted by the prefixes M, B, and C in Table 1 and elsewhere in this paper. The Island Intrusion samples (G1 and G2) are from area C (Figs. 1, 2). In each area samples were obtained from several sites, and at each site four to eight oriented cores were drilled. Each core was then sliced, yielding two cylindrical rock specimens.

The M sites are from the mountainous region west of Buttle Lake. Sites M1–M9 and M14 are in two separate sills intruding limestones of the Buttle

TABLE 1. Results from each collecting site

Site	Grid reference	DDD°, dip°	$N(R)k$	$D°, I°$	$\lambda°, \phi°$	Treatment
B01(X)	92F12/175/041	005, 25	6(5.88)041	016, −31	22, 038E	20–30 mT, 550°C
B02(X)	92F12/174/042	005, 25	8(7.89)061	001, −33	22, 054E	25–30 mT
B03(X)	92F12/168/047	005, 20	4(3.87)022	022, −35	21, 053E	20–30 mT
B03(Y)	92F12/168/047	—	3(2.72)007	344, +79	69, 142W	20–30 mT, 500°C
B04(X)	92F12/166/063	010, 25	4(3.91)033	006, −27	26, 048E	10–30 mT
B04(Y)	92F12/166/063	—	4(3.81)016	005, +55	75, 038E	20–50 mT, 550°C
B05(X)	92F13/125/349	065, 12	6(5.86)038	359, −33	22, 056E	40 mT
B06(Y)	92F13/125/349	—	3(2.98)113	004, +71	84, 103W	20 mT
B07(Y)	92F13/125/349	—	4(3.94)046	045, +70	62, 061W	20 mT
B08(X)	92F13/124/349	065, 12	5(4.76)017	350, −21	29, 066E	40 mT
B09(X)	92F13/123/349	065, 12	6(5.92)062	330, −22	23, 087E	20–50 mT, 550°C
B10(X)	92F13/120/344	075, 12	4(3.76)013	337, −24	24, 079E	30 mT
B11(X)	92F13/119/348	075, 12	5(4.91)046	345, −12	33, 072E	20 mT
B12(X)	92F13/097/316	060, 12	6(5.92)064	350, −25	26, 066E	30 mT, 550°C
B13(X)	92F13/097/315	060, 12	6(5.98)229	002, −30	24, 053E	40 mT, 550°C
B14(X)	92F13/096/313	060, 12	8(7.86)050	002, −42	16, 053E	20–40 mT, 600°C
B15(X)	92F13/096/312	060, 12	3(2.77)009	021, −36	18, 034E	20–40 mT, 550°C
B15(Y)	92F13/096/312	—	2(1.99)	022, +58	71, 008W	20–30 mT, 550°C
M01(X)	92F12/084/081	287, 25	7(6.89)056	007, −41	17, 048E	20 mT, 550°C
M02(Y)	92F12/084/081	—	5(4.97)143	012, +63	80, 007W	10 mT, 550°C
M03(Y)	92F12/084/081	—	6(5.94)080	005, +42	64, 056E	10 mT, 550°C
M04(Y)	92F12/085/078	—	6(5.92)060	031, +42	55, 000	20 mT, 550°C
M05(X)	92F12/085/078	287, 25	4(3.95)063	359, −37	20, 056E	30–40 mT, 550°C
M06(X)	92F12/084/079	287, 25	1	026, −24	24, 027E	15 mT, 550°C
M06(Y)	92F12/084/079	—	4(3.65)009	045, +44	48, 017W	15–40 mT
M07(Y)	92F12/084/079	—	6(5.88)043	054, +71	58, 066W	20 mT, 550°C
M08(X)	92F12/077/078	280, 22	5(4.69)013	035, −19	23, 017E	30 mT, 550°C
M09(X)	92F12/077/078	280, 22	5(4.93)066	004, −34	22, 051E	30 mT, 550°C
M10(X)	92F12/075/080	280, 22	6(5.90)049	023, −35	18, 032E	20 mT, 550°C
M11(X)	92F12/125/067	300, 26	6(5.58)012	029, −29	20, 025E	20–30 mT, 600°C
M12(X)	92F12/124/067	300, 26	8(7.84)045	010, −42	16, 045E	30 mT, 550°C
M13(X)	92F12/124/067	300, 26	5(4.88)034	003, −31	24, 052E	20–30 mT, 550°C
M14(X)	92F12/076/080	280, 22	5(4.79)019	018, −30	22, 036E	20–30 mT 500–550°C
C01(X)	92K3/356/488	015, 12	5(4.85)027	041, −33	13, 015E	40–50 mT, 550–650°C
C02(X)	92K4/239/597	065, 10	5(4.93)056	010, −47	11, 045E	30–60 mT, 550–650°C
C03(X)	92K4/193/639	030, 15	5(4.84)027	006, −58	01, 050E	30–50 mT, 550°C
C04(Y)	92K4/192/647	—	5(4.94)073	021, +45	61, 012E	20 mT, 500°C
C05(Y)	92K4/175/678	—	5(4.66)012	046, +71	62, 064W	20 mT, 500°C
C06(Y)	92K4/184/658	—	7(6.64)017	028, +34	51, 009E	15–20 mT
C07(Y)	92K3/310/587	—	6(5.56)011	036, +25	43, 003E	15–20 mT, 550°C
C08(X)	92K3/307/589	170, 10	5(4.88)032	037, −28	17, 017E	80–100 mT, 650°C
C09(X)	92K3/307/589	170, 10	6(5.60)013	025, −36	16, 030E	80–100 mT, 650°C
G01	92K5/125/757	—	5(4.95)084	324, +81	63, 149W	10 mT, 400°C
G02	92K5/115/732	—	5(4.93)061	205, +67	12, 142W	10–15 mT, 500°C

NOTES: The site number is given on the left. B, M, and C denote the Buttle Lake, Marble Lake Peak, and Campbell River sampling areas (see Figs. 1, 2). G01 is in an Island Intrusion and G02 its contact aureole. X and Y are types of magnetizations. DDD°, dip° column gives the azimuth of the downdip direction and the magnitude of the geological dip. Column headed $N(R)k$ lists the number of oriented cores (usually two specimens measured per core), the resultant, and Fisher's estimate of precision. D, I are the mean declination and inclination and λ, ϕ the corresponding pole position. The X directions are corrected for the geological attitude as given in the third column. The Y directions are given with respect to present horizontal.

Lake Formation, the uppermost unit in the Sicker Group (Yole 1965, 1969), on the ridge trending south from Mt. McBride. M10 is in thin-bedded siliceous sedimentary rocks overlying the upper sill and probably equivalent to the argillite unit which elsewhere contains Middle Triassic (Ladinian) fossils (Carlisle 1972; Muller 1977*b*). Sites M11–M13 are from a sill in the upper part of the Buttle Lake Formation below Marble Peak. Reliable attitudes for the M sites are available from the adjacent well-bedded sedimentary rocks, which were deformed after intrusion of the diabase.

Sites B1–B4 are from diabase or massive basalts near lake level on the east side of Buttle Lake. The samples are from the basal part of the Karmutsen, where local attitudes are obtainable from bedded rocks of the immediately underlying Buttle Lake Formation. Sites B5–B15 are from the east side of Upper Campbell Lake, where amygdaloidal lavas and massive basalts within the upper part of the

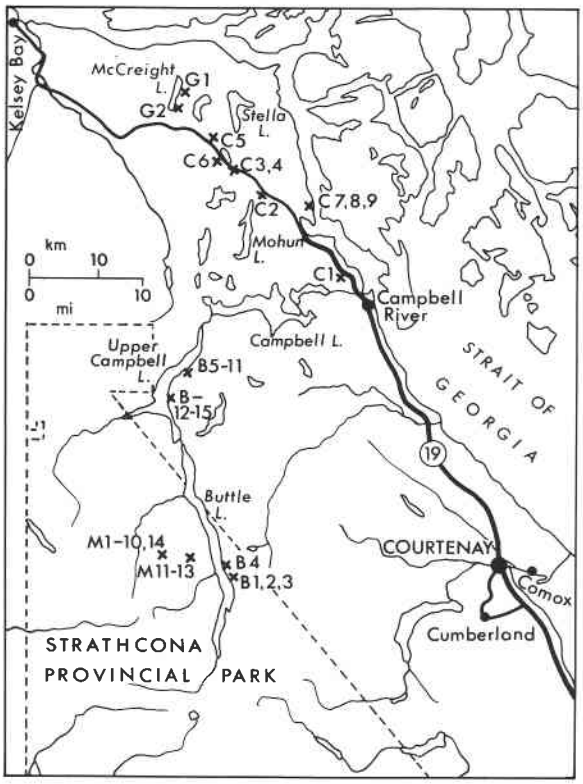


FIG. 2. Locations of sampling sites.

Karmutsen are exposed. Reliable local attitudes are difficult to obtain for the sites themselves, but regional attitudes within the major fault block containing them consistently dip northeast at low to moderate angles. Sites B5–B11 are close to the contact zone of a small body of Island Intrusions. Sites B12–B15 are about 1.5 km from the igneous contact. Although no local structural disturbance is evident in sites B5–B15, the possibility of local attitudes divergent from those of the major block is not disproven, and as we shall show below, there is some evidence in favour of this at sites B5–B11.

The C group of sites extends north and northwest of the town of Campbell River. Sites C1–C6 are along provincial highway 19, between Campbell River and Kelsey Bay, and sites C7–C9 are on the south shore of Brown Bay, at the north end of Seymour Narrows. The rocks are massive to amygdaloidal basalts. All are from the upper and middle parts of the Karmutsen, and all are in the prehnite–pumpellyite zone of burial metamorphism (Kuniyoshi 1972; Kuniyoshi and Liou 1976*a*, *b*). Regional dips are mainly to the northeast, but local variations occur at Brown Bay and are probably due to proximity to major faults.

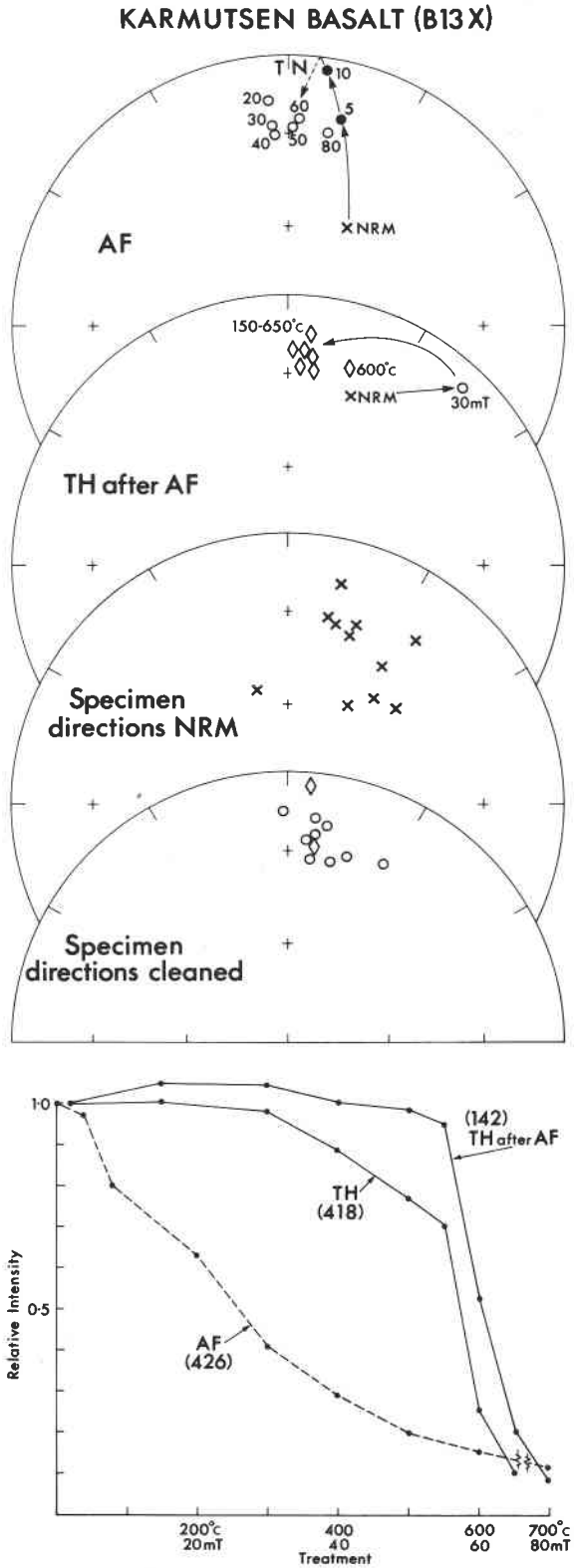
The samples collected represent units extending through much of the stratigraphic range of the Karmutsen Formation, and include several diabase sills inferred to be co-magmatic with the lavas of the formation. Reliable local attitudes are available for most of the sites, including some from each sampling area. Regional dips within major fault blocks are relatively consistent, generally less than 30°, thus enabling reasonable estimates for attitudes in the remainder of the sites, in the absence of local evidence of strong structural disturbance. The diabase sills sampled are conformable with the intruded layered rocks and were folded or tilted in fault blocks along with the country rocks after intrusion.

X Magnetization

As in our previous study, we find that two families of magnetization directions are present, *X* and *Y*. The *X* magnetization has negative (upward) inclination and declinations ranging from north-northeast–north-northwest. It is hard magnetically and characterizes the Karmutsen Formation because it occurs throughout its thickness. Typical behaviour is illustrated in Fig. 3. The natural remanent magnetization (NRM) is directed toward the north-northeast, with positive inclination lower than that of the present geomagnetic field. Both alternating-field (AF) and thermal treatments in near-zero residual field (apparatus described by Roy *et al.* (1972, 1973)) cause the directions to become northerly with shallow negative inclinations, and these directions remain little changed in AF up to 100 mT (1 mT = 10 Oe) and temperatures of 600°C. The thermal decay curves are square-shouldered, especially if the specimens are first treated in AF of a few tens of milliteslas to remove magnetizations with low coercivities. The *X* magnetization is characterized by unblocking fields up to 100 mT, and unblocking temperatures in a narrow range of less than 100°C between 500 and 700°C. Other examples of *X* magnetization have been documented previously (Irving and Yole 1972, p. 92).

Magnetizations with low unblocking temperature (generally less than 400°C and always less than 500°C) and low unblocking fields (generally less than 30 mT) are removed during demagnetization and vector subtraction shows that they have directions in the northeast quadrant, generally close to the present geomagnetic field, but which are more scattered than would be expected from the removal of VRM caused entirely by the present field. This could mean that the removed magnetizations are, entirely or in part, a *Y* magnetization (see below).

A hard *X* magnetization of high directional sta-



bility is found in sediments of the unnamed unit at site M10, 7 m above the contact with a 100 m thick sill. These rocks were undoubtedly heated during intrusion of the sill. Their directions are statistically indistinguishable from those in the underlying diabase (M8, M9, M14). This constitutes a partial contact test. To be fully effective the test should also demonstrate that rocks into which the Karmutsen sills have been intruded, and which are situated some distance from the sills outside the zone of remagnetization, possess different directions. In our attempt to complete this test we have sampled the Sicker Group far removed from contacts, but have not yet found directionally stable magnetizations. Therefore, the results from site M10, although they strongly suggest that X magnetization is original and was acquired at the time of intrusion, are not, taken in isolation, proof of this hypothesis.

X magnetizations have been observed at 28 sites. The mean directions are plotted in Fig. 4 and listed in Table 1, corrected for the geological tilts given in the third column of Table 1. Fisher's precision of the site mean directions increases from 18 to 21 when corrected for tilt, but the change is not significant because the tilts are small.

Sites B5 and B8–B11 have declinations systematically west of the main group of X directions (Fig. 4, Table 1). These sites are from the northern part of sampling area B, high in the sequence, and are referred to as the B subset (Table 2, entries 15 and 16). The B subset has hardness characteristics common to all other X magnetizations. The B subset has a mean declination of $344 \pm 11°$, and the X magnetization, excluding the B subset, has a declination of $015 \pm 6°$. Therefore the difference, using the relationships set out in the Appendix, is $31 \pm 13°$, which is significant at $P = 0.05$. This deviation is unlikely to be caused by unidentified magnetic overprints because the X magnetizations at these sites are univectorial and are amongst the hardest

FIG. 3. Typical hardness properties of the X magnetization. The upper two stereograms show directions observed in specimens from site B13 after successive AF (circles) and thermal (TH) (diamonds) treatment. AF peak strengths are given in millitesla (1 mT = 10 Oe). The lower two stereograms show all specimen directions from site B13 before and after cleaning. North-seeking directions are plotted; open symbols denote negative (upward) inclination and solid symbols positive (downward) inclination; the perimeter is the present-day horizontal, the directions being uncorrected for geological tilt. The point labelled NRM is the direction of natural remanent magnetization. Normalized decay curves are given below, on which the initial intensities of magnetization are given in 10^{-3} A/m (1 A/m = 10^{-3} G).

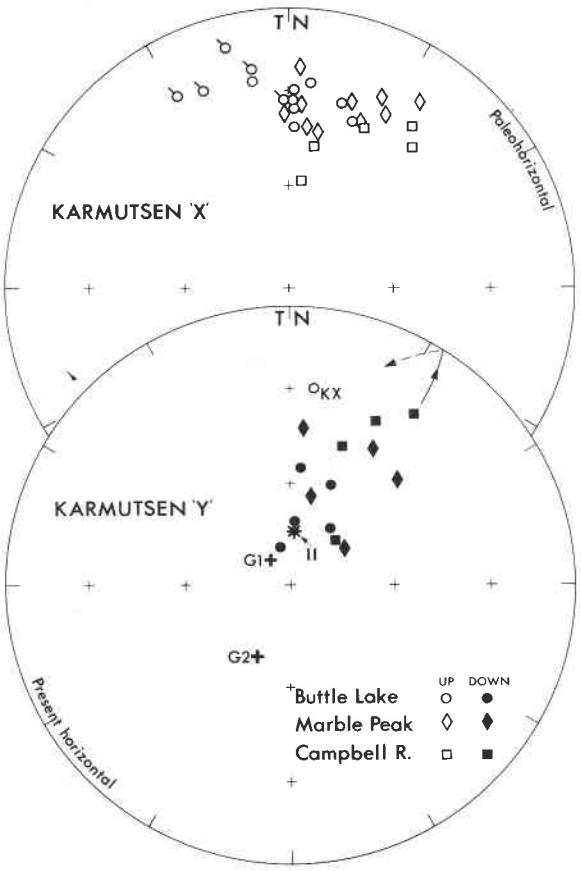


FIG. 4. Site mean directions for *X* magnetizations are shown above, corrected for geological tilt. The B subset is marked by flecked circles (see text). *Y* magnetization directions (uncorrected) are shown below. Open (solid) symbols denote negative (positive) inclinations. Legend shows symbols for each of: Buttle Lake (B), Marble Peak (M), and Campbell River (C) groups of sites (Table 1, Fig. 2). G1 is the direction observed by us at one site in the Island Intrusions and G2 is in the metamorphic aureole. II is the mean direction observed by Symons (1971*a*) from 17 sites in the Island Intrusions. KX is the mean direction of the *X* magnetization.

we have observed from the Karmutsen. When corrections are made for inferred dips the directions do not move into the main group. We suggest two possible explanations for this result: firstly, a local anticlockwise rotation (not evident geologically) of the subset B sites, possibly related to the close proximity of an Island Intrusion and secondly, a temporal excursion of the field. Northwest directions were also observed by Symons (1971*a*, Fig. 4) and Irving and Yole (1972). In the analysis, averages are given including and excluding the B subset (Table 2). The effect of subset B on the overall mean value is small.

Schwarz *et al.* (1980) have also observed magnetizations with properties and directions like our *X* group, and, like us, they recognize a spread in declination from north-northeast–north-northwest. They have subdivided their results into two groups ("northwest-up," 4 sites and "north-up," 4 sites, Table 2, entries 3 and 5). These groups may be comparable to our B subset and *X* magnetizations, excluding the B subset, respectively (Table 2, entries 15 and 17). Schwarz *et al.* (1980, pp. 395, 396 and Fig. 6) report that certain high field experiments reveal differences between their two groups. However, the characteristics of the natural remanent magnetization of the two groups are not noticeably different—unblocking temperatures generally between 550 and 600°C and unblocking fields exceeding 80 mT (Schwarz *et al.* 1980, Figs. 2, 3). The mean inclinations of their two groups are statistically indistinguishable, as we also find for *X* directions. Likewise, the groups with the more westerly declinations are from the upper part of the sequence. The general similarities in the remanence characteristics described in both studies suggests that the declination spreads observed have a common origin. We suggest that either local tectonic rotations or a short-term excursion of the field are possible causes, the former being preferred at the present stage of investigation.

Y Magnetizations

The *Y* magnetizations have site directions that mainly fall in the northeast quadrant, with one value directed toward the north-northwest (Fig. 4). The properties of *Y* magnetizations are heterogenous and we shall argue that they were acquired at different times by different processes. Like the *X* magnetization, the *Y* magnetizations commonly remain directionally stable, either through the full demagnetization range, or after removal of low stability components, but their coercivities and unblocking temperature spectra are much more variable.

In Fig. 5, *Y* magnetizations having low unblocking fields and temperatures are illustrated. At this site *X* magnetizations also occurred, but in different specimens. In those specimens having *Y* magnetization there is no systematic change in direction throughout the range of both AF and thermal demagnetization, but there is considerable scatter from specimen to specimen. The intensities fall rapidly during demagnetization. Evidently only one ill-defined magnetization is present. It cannot be a VRM caused by the present field because its inclination is too shallow. All the *Y* magnetizations from the Buttle Lake and Upper Campbell Lake

TABLE 2. Summary of directions and poles from the Karmutsen Formation

	B		k	A_{95}
1. X magnetization paleopole (1)	5	13N, 45E	—	23*
2. X magnetization paleopole (2)	3	21N, 61E	—	17*
3. Northwest-up subset, X directions (4)	4	327, −25	41	15
4. Northwest-up subset, X paleopole (4)	4	21N, 89E	—	12*
5. North-up subset, X directions (4)	4	000, −32	65	12
6. North-up subset, X paleopole (4)	4	23N, 55E	—	4*
7. Buttle Lake X directions (5)	13	356, −29	30	08
8. Buttle Lake X VGP's (5)	13	24N, 58E	31	08
9. Marble Peak X directions (5)	10	016, −33	40	08
10. Marble Peak X VGP's (5)	10	21N, 39E	41	08
11. Campbell River X directions (5)	5	026, −41	24	16
12. Campbell River X VGP's (5)	5	12N, 11E	23	16
13. Mean of X directions (5)	28	008, −33	21	06
14. Mean of X VGP's (5)	28	21N, 44E	21	06
15. B subset (B5, B8–B11) X directions (5)	5	344, −23	41	12
16. B subset (B5, B8–B11) X paleopoles (5)	5	26N, 72E	51	11
17. X directions excluding B subset (5)	23	015, −35	33	05
18. X paleopoles excluding B subset (5)	23	20N, 41E	32	05
19. Grand mean X directions 3 + 5 + 13	36	003, −33	17	6
20. Grand mean X paleopole, from 19	36	23N, 52E	—	6*
21. Y magnetization paleopole (1)	7	82N, 48E	—	19*
22. Y magnetization paleopole (2)	8	70N, 155W	—	30*
23. Y magnetization paleopole (4)	4	52N, 22E	—	22*
24. Mean of Y directions (5)	14	025, 57	18	10
25. Mean of Y VGP's (5)	14	70N, 15W	13	11
26. Expected Late Triassic paleopole (3)	9	68N, 093E	—	06

NOTES: B is the number of sites, each given unit weight, k is Fisher's estimate of precision of directions or VGP's, and A_{95} is the circle of confidence ($P = 0.05$). Sources for data: (1) Symons (1971*b*); (2) Irving and Yole (1972); (3) Irving (1979*b*); (4) Schwarz *et al.* (1980); (5) this work. X magnetization directions corrected for tilt. Y directions quoted with respect to horizontal.
*Estimate obtained by averaging *dm* and *dp* values.

sites (B area) are of this type. Another example of similar behaviour is given in Irving and Yole (1972, p. 92).

However, Y magnetizations in many samples from the A and C sampling areas show different properties, and results from three specimens are shown in Fig. 6. The NRM directions are widely scattered from specimen to specimen. Thermal or AF cleaning yield consistent Y directions. Alternating-field followed by thermal cleaning also yields the Y direction with high unblocking temperatures between 550 and 600°C. The thermal decay curve obtained without prior AF demagnetization has a sharp minimum at 300°C, showing that a magnetization with a direction strongly divergent from that of Y is being removed at low temperatures. The specimen demagnetized in AF gave a curve that at first falls rapidly as the directions migrate to the Y direction and then remains little changed up to 100 mT. Evidently, in all three specimens magnetizations with either low unblocking temperatures or fields and with widely varying directions are present. No systematic study of these soft magnetizations has been made; we have been content to isolate the more stable Y magnetization.

The Y magnetizations were observed at 14 sites (Fig. 4). The site mean directions have an oval distribution, spreading from a direction not far from the present field towards the mean direction of the X magnetization. In Fig. 4 and Table 1 the directions are given with respect to the present horizontal. The directions are somewhat more dispersed after correction for tilt but the change is not significant; Fisher's precision of the site means changes from 18 before to 14 after correction. However, no realistic corrections for tilt can be made because, as we shall argue later, the age of the Y magnetization is known only within wide limits, and age of the tilting also is not precisely known and could have been spaced over a considerable interval of time spanning the period over which the Y magnetizations were acquired. The effect of tilting on the directions can only have been slight, however.

Schwarz *et al.* (1980) describe magnetizations at four sites ("north-northeast-down" group) with hardness characteristics and directions similar to our Y magnetization (cf. our Fig. 6 with specimen 53, Fig. 3 of Schwarz *et al.* (1980)).

Discussion of Magnetizations

In most of our specimens the magnetizations were predominantly either X or Y. Although the softer magnetizations removed during the uncov-

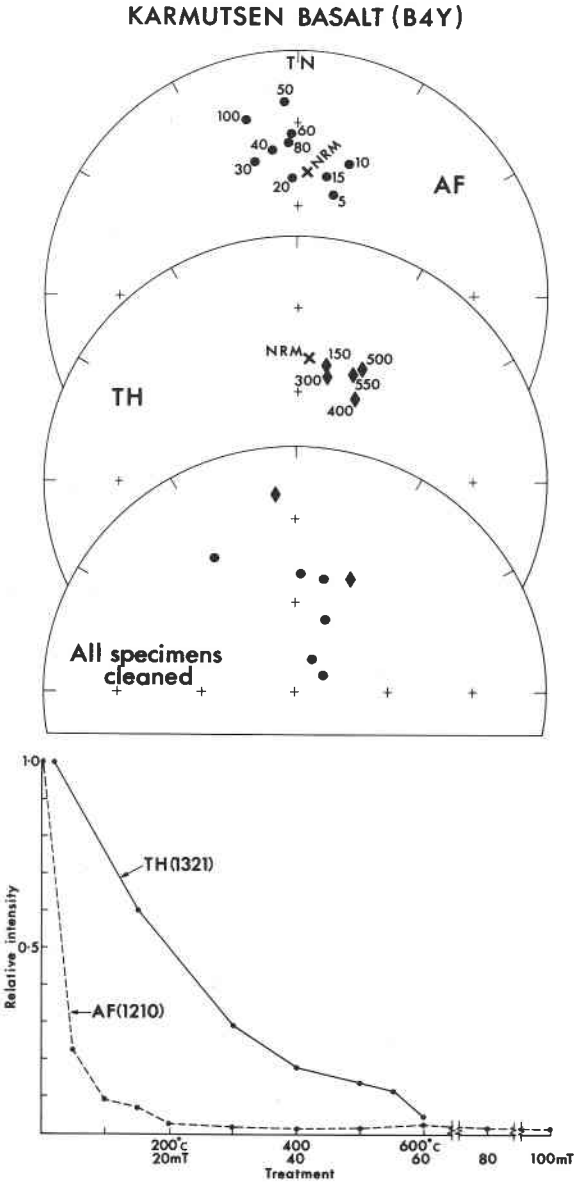


FIG. 5. Magnetically soft *Y* directions (site B4). The stereograms show directions of magnetization of a single specimen during AF treatment (top) during thermal (TH) treatment (middle), and all specimens after cleaning (bottom). Decay curves below correspond to the upper two stereograms, on which initial intensities are given in 10^{-3} A/m.

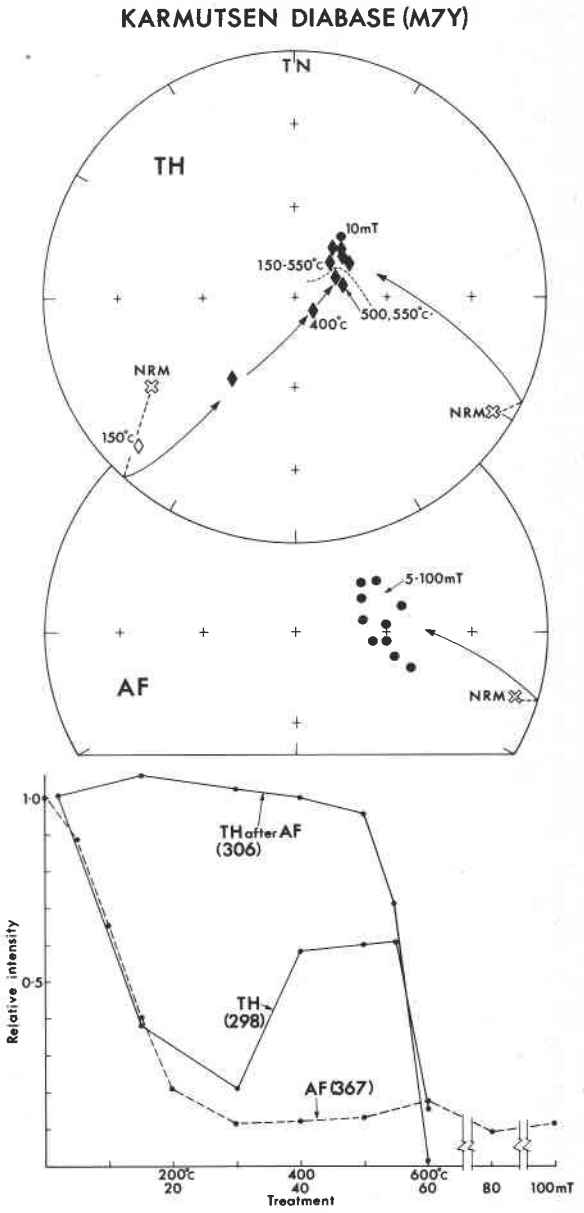


FIG. 6. Magnetically hard *Y* directions (site M7). The stereograms show the directions of magnetization in individual specimens during stepwise thermal (TH) and AF treatment. In the upper diagram the specimen on the left has been subjected to thermal treatment only, and the specimen on the right has first been cleaned at 10 mT and afterwards thermally treated. In the centre a third specimen has been subjected to AF's only. The corresponding decay curves are shown below, with the initial intensities in 10^{-3} A/m.

ering of *X* magnetizations could be *Y* magnetizations, we have found no unequivocal evidence of overprinting of *Y* on *X*. Both *X* and *Y* magnetizations occur in diabase sills, amygdaloidal and massive basalt flows. *Y* was found also in pillow basalts of our first collection (Irving and Yole 1972). At site B4, where both magnetizations occur, the *X* magnetization is found in relatively fresh blue–grey basaltic rocks with strong magnetic intensities, whereas the *Y* samples are from weakly magnetized, greenish, altered basalts. Such a correlation of magnetic and petrographic properties has been observed at several sites, but the correlation is not clearly demonstrable across the entire collection.

The summary results are listed in Table 2. We believe that the *X* magnetization (entries 13 and 14) is the original magnetization acquired at the time of formation because it is found throughout the great thickness of the Karmutsen, because it is characterized by high unblocking temperatures and fields, and because the contact test, insofar as it has as yet been possible to carry it out, is positive. Only one polarity is present. The paleopole labelled K in Fig. 7 is the pole toward which the directions (north-seeking) of the *X* magnetization point. If the field was then of normal polarity, K is the geographic north pole. Intervals of normal polarity are more frequent in the Late Triassic than intervals of reversed polarity (Irving and Pullaiah 1976) and this provides some basis for arguing that K actually is the Late Triassic north pole. The evidence, however, is not conclusive because the time spanned by the Karmutsen is insufficient to provide a representative sample of the polarity spectrum of the Late Triassic geomagnetic field. The paleopole (Table 2, entry 20) calculated from the grand mean of *X* directions of 23°N,52°E ($A_{95} = 6°$), is in excellent agreement with our original determination (21°N,61°E, $A_{95} = 17°$, Irving and Yole 1972).

The results of Hillhouse (1977) from the Nikolai Greenstone are also plotted on Fig. 7. Both of the two possible paleopoles are plotted (N1 and N2), and either of them may correspond to K. We suggest that N1 corresponds to K because this assumption provides a pattern of magnetizations more consistent with other data from the westernmost Cordillera. We give three reasons: (1) most (90%) of the north-seeking directions of the Nikolai point to N1; (2) N1 is right-handed like all the Karmutsen poles (Fig. 7); (3) the rotation implied by N1 is not significantly different from that required by the Jurassic pole AJ obtained from the nearby SAJ terrane in southern Alaska (Packer and Stone 1974), whereas the rotations implied by N2 are grossly different as can be seen in Fig. 7.

The *Y* magnetizations are, in general, less hard than *X*, the majority markedly so. *Y* directions are scattered, but they are by no means random. Their diverse properties are not distributed uniformly through the Karmutsen Formation. Sample area B tends to have softer *Y* magnetizations than the other areas, whose virtual geomagnetic poles (VGP) tend to be more far-sided (Fig. 7). An important feature of the *Y* magnetization VGP's is that they fall in the same general area as the aberrant poles from Jurassic and Cretaceous rocks of the western Cordillera (Fig. 7). These properties are best explained by supposing that the *Y* magnetizations are secondary, and that they were probably

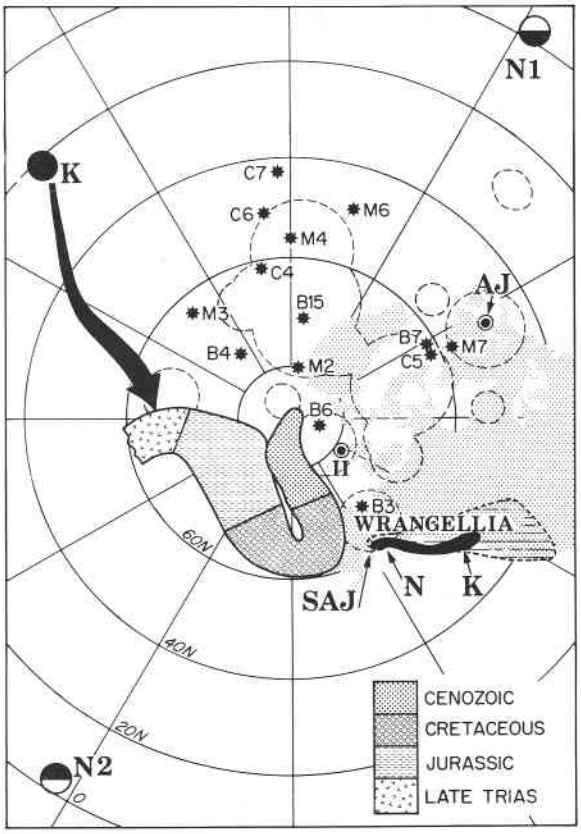


FIG. 7. The pole positions from Wrangellia compared with the path of apparent polar wander for North America. The poles N and K are calculated from what is considered to be the original magnetization of the Nikolai Greenstone (Hillhouse 1977) and the Karmutsen Formation (*X* magnetization, entry 8, Table 2). For the Nikolai, both poles are plotted (N1 and N2). The large arrow points to the paleopole that the Karmutsen would have given if Vancouver Island had not moved relative to North America. The small stars are VGP's calculated from the *Y* magnetization at individual sites (Table 1). The lightly-dashed envelopes are the error circles of the aberrant Jurassic and Cretaceous paleopoles obtained from the western Cordillera sampling area shown by horizontal shading (compiled from Irving (1979*b*)). Paleopole II is by Symons (1971*a*) from the Island Intrusions. Present position of Wrangellia and sampling areas for Nikolai and Karmutsen Formations also shown. AJ is the paleopole from the southern Alaskan terrane SAJ.

acquired at various times during the Jurassic or Cretaceous, during deep burial when the subgreenschist facies metamorphism brought about chemical changes in the magnetic minerals (CRM), during regional heating at the time the Island Intrusives were emplaced (viscous partial TRM), or during final uplift (viscous partial TRM). It is presumably because of their varied origins and widely spread ages of acquisition that the *Y* magnetizations have such heterogenous properties. On the other hand, one could argue that the oval distribu-

tion of *Y* directions (Fig. 4) indicates that they are not true field directions but contain underlying *X* magnetizations; we have found no evidence for this.

The Karmutsen is intruded by granitoid plutons, the Island Intrusions, of Jurassic age. The original magnetization of the Island Intrusions, if it can be identified, should show significantly different directions from those of the Karmutsen *X* magnetization if the latter is indeed the original Late Triassic magnetization. In the extensive survey by Symons (1971*a*), a steeply inclined positive (downward) magnetization was found (II of Figs. 4, 7, 8). For comparison, we have sampled an Island Intrusion pluton at site G1 (Fig. 2). Thermal and AF demagnetization show a rapid drop in intensities, the mean destructive field being very low, about 5 mT, and the unblocking temperatures are less than 400°C. The directions during demagnetization are somewhat scattered, but the average direction is close to that observed by Symons (Figs. 4, 8). These characteristics are typical of VRM or viscous partial TRM and thus are unlikely to represent the original directions acquired during emplacement. As can be seen in Fig. 4, the directions in the granodiorite are close to some of the *Y* directions of the Karmutsen, and the granodiorite magnetization has other properties similar to those of *Y* magnetizations, particularly the *Y* magnetizations observed in sampling area B (Fig. 5).

We have sampled Karmutsen in the contact aureole of the same pluton (G2, Fig. 2). The rock has reached amphibolite grade of metamorphism (Kuniyoshi and Liou 1976*a*). No corrections for possible tilting could be applied. The magnetization is directionally stable, with directions differing from those in the pluton (G1, Fig. 4) and from any direction yet observed in the Karmutsen. This result is included for the record and we have no interpretation for it. It serves to illustrate how complex the magnetizations of the Island Intrusions and their aureoles are likely to be, and why they cannot, at present, be used to place constraints on the age of the magnetizations observed in the Karmutsen.

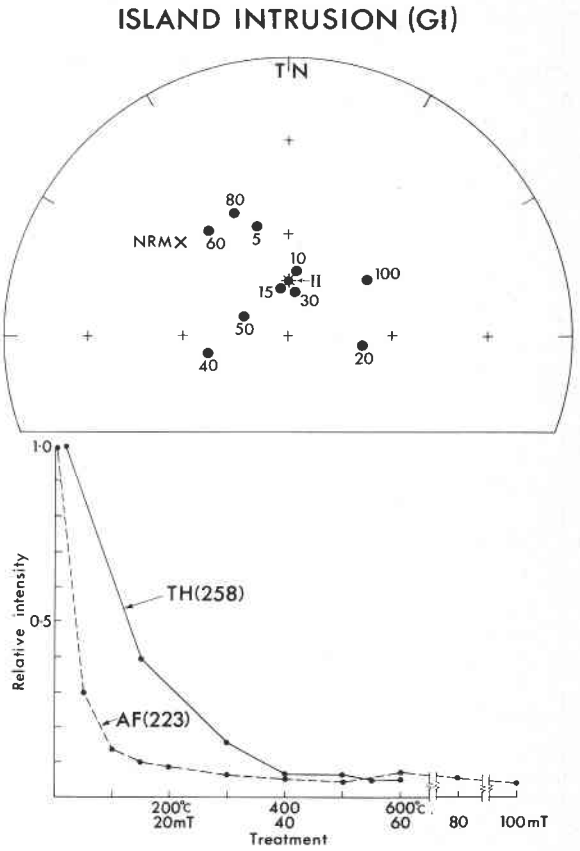


FIG. 8. Demagnetization of granodiorite from an Island Intrusion (site G1). Directions changed during AF demagnetization are shown above with the corresponding decay curves below. II is the mean direction from 17 sites in Island Intrusions (Symons 1971*a*).

Tectonic Applications

The *X* magnetization is, we believe, very probably the original Late Triassic magnetization indicative of the geomagnetic field at that time. The corresponding paleopole is far-sided and right-handed with respect to the North American APW path (K of Fig. 7).

The *Y* magnetizations yield VGP's that, with two exceptions (B3 and B6), are significantly different, both individually and collectively, from the APW path for North America (Fig. 7, Table 2). Thus the *X* magnetization and most of the *Y* magnetizations were acquired when Vancouver Island did not have its present position relative to cratonic North America. This is a general argument, independent of any assumptions about the relative or absolute ages of the *X* and *Y* magnetizations. Except for B3 and B6, *all* the magnetizations we have observed from the Karmutsen of Vancouver Island diverge by large angles from the direction anticipated from cratonic North America. Evidently Karmutsen rocks, like many others from western Cordilleran localities (summarized in Irving 1979*b*), belong to crustal blocks that have moved independently of their coeval counterparts in the stable interior of North America.

Aberrant paleopoles could be the result of one or more of the following: (*a*) tilting about a local horizontal axis; (*b*) relative paleolatitudinal displacement, RPD (translation with reference to North

America); (*c*) rotation about a local vertical axis, RR. In layered rocks the effects of postmagnetization tilting (*a*) can be removed. If after correction the paleopoles remain aberrant, then translations (*b*) and (or) rotations (*c*) of the host rock have occurred and can be calculated as shown in the Appendix.

It is not possible to define uniquely the real motions that give rise to an aberrant paleopole, and this may be illustrated by considering translations. (See Appendix for details.) Translations can be either orthogonal (parallel to lines of latitude or longitude) or oblique. Motion parallel to latitude cannot be detected paleomagnetically. Motions parallel to longitude cause changes in paleolatitude that are mapped by either far-sided or near-sided poles. Rotations produce left-handed or right-handed poles. Oblique translation would produce both signatures. For example, oblique translation towards western North America from the southwest (northwest) produces far-sided (near-sided) and right-handed (left-handed) poles. The former is the distinctive pattern of paleopoles from the western Cordillera (Fig. 7) and could, therefore, have been produced either by northward translation coupled with clockwise rotation, or by oblique convergence of one or more aberrant terranes from the southwest. The northward component of motion remains the same. It is noteworthy that oblique convergence has earlier been suggested as a means of accreting Cordilleran terranes to cratonic North America (Monger and Price 1979, p. 782; Souther 1977, p. 10; Paterson 1977, p. 1338). Detailed review and discussion of the geological evidence for Cordilleran evolution are beyond the scope of this paper; we shall dwell only on the significance of the paleolatitude results from the Insular Belt.

The mean inclination of the *X* magnetization we have observed is 33 ± 6° corresponding to a paleolatitude of 18 ± 6°. The mean inclinations of the two *X* magnetization subsets observed by Schwarz *et al.* (1980) are 25 ± 15° and 32 ± 12°, yielding paleolatitudes of 13 ± 15° and 17 ± 12°, in general agreement with our result. Our result gives a paleolatitude which is either 13° or 49° further south than the paleolatitude expected for Vancouver Island, depending on whether the paleolatitude determined from *X* is taken to be north or south of the paleoequator. In Fig. 9, possible positions relative to the Late Triassic reassembly of the Americas are represented by north and south latitudinal belts (stippled) centred at either 18°N, which is comparable to southwestern United States, or at 18°S latitude, which is comparable to that of northwestern South America at that time.

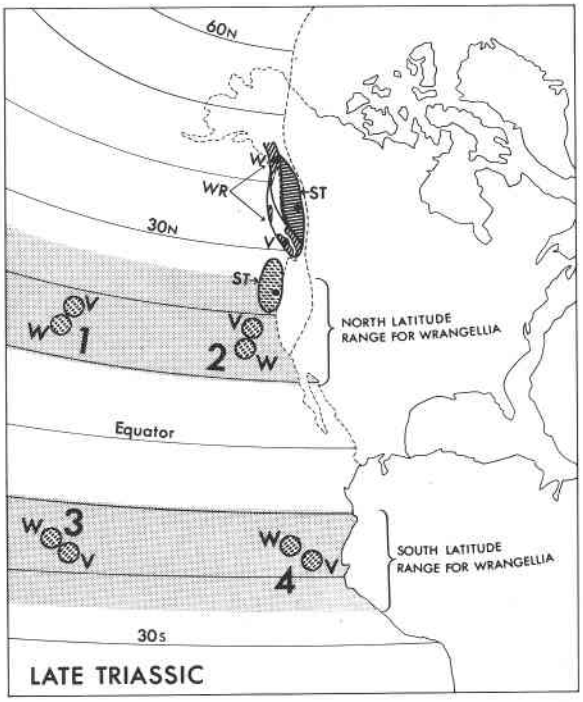


FIG. 9. Reconstructed positions for Wrangellia and the Stikine terrane for Late Triassic, based on paleomagnetic results reported herein and in Monger and Irving (in press) and Hillhouse (1977). Present positions shown in solid bars (WR and ST). Preferred Late Triassic position for the Stikine terrane and four of the possible positions for Wrangellian elements (W, Wrangell Mountains; V, Vancouver Island) shown in broken-bar ornament within latitudinal belts (stippled) bounded by the paleolatitude error limits. The relative positions of W and V are also not certain; whether V or W was in "inboard" position in Late Triassic is still conjectural.

Positions 1, 2, 3, and 4 of Fig. 9 represent some possible positions for Wrangellia (Vancouver Island, the Queen Charlotte Islands, and the Wrangell Mountains of Alaska), determined from our results and those of Hillhouse (1977). The sideways extension of the belts indicate the indeterminacy in longitude. Irving and Yole (1972) favoured the southerly option, preferring the greater translation and the lesser rotation, that is, a translation of about 5000 km and a clockwise rotation of 22° of Vancouver Island (see Appendix). In the southerly option, the paleopole K of Fig. 7 is assumed to be the geographic north pole, and the internally consistent pattern referred to above is maintained. Muller (1977*b*) favours a northern option, deriving Wrangellia from offshore "California." Hillhouse (1977) also favours the northerly option, preferring the lesser translation and the greater rotation, that is, a translation of about 3000 km and an anticlockwise rotation of about 79° for the Wrangell Mountains. In this northerly op-

tion, K (Fig. 7) is assumed to be the south pole and N2 the north, and the pattern is no longer consistent; the dominant polarities no longer point to the same pole and the rotations of different parts of Wrangellia are no longer all clockwise.

A similar problem arises for southern Alaska (SAJ of Fig. 7), and, because some of the rocks (the Nabesna sequence) that have been studied by Packer and Stone (1974) may overlie Wrangellian terrane, the recent discussion by Stone (1979) is of relevance here. He argues that the southerly latitude option is to be preferred because it is in good accord with the plate tectonic reconstructions of the Pacific Ocean given by Hilde *et al.* (1976), based on the independent evidence of magnetic anomalies, and because the inferred motion of SAJ is then continuously northward. On the other hand, if the northerly option is taken for Wrangellia and the southern Alaskan terrane, there is a sharp change in direction of motion of SAJ in the Cretaceous from south–north to north–south (Stone and Packer 1977, 1979). The possibility of such an abrupt change cannot be discounted, but the assumption of continuity is reasonable and provides some indirect support for the southerly option for Wrangellia (Fig. 9). The southerly option requires that Wrangellia traverse the equator en route to its present position, but a search of the literature yields no conclusive paleoclimatic record of such an equatorial passage. Definitive paleomagnetic results should be obtainable from Jurassic rocks on Vancouver Island and this is intended to form the next stage of our study.

Nur and Ben-Avraham (1978) have suggested that various crustal elements, including Wrangellia, that are now embedded in orogenic belts bordering the Pacific Ocean were derived from a single Late Paleozoic continent which they term "Pacifica." Pacifica was attached to Australia and Antarctica forming an integral part of Gondwana, and was subsequently dismembered and dispersed by sea-floor spreading in the Pacific. In Fig. 10 the Triassic paleolatitudes of Gondwana and Vancouver Island are compared. That part of Pacifica (B of Fig. 10) from which Wrangellia was supposedly derived was then at 50–80°S. According to Nur and Ben-Avraham (1978, Fig. 2), Pacifica had begun to break up by the Late Triassic and the paleolatitude may have, by then, decreased to 40°S. These speculations are inconsistent with evidence from Wrangellia on two counts. Firstly, geological evidence indicates that in the Paleozoic and Triassic Wrangellia consists of arc and rift terranes, not continental crust. Secondly, the paleolatitudes of Vancouver Island (18°) and the Wrangell terrane (15°) are not consistent with a position close to the south end of Pangea during the Late Triassic. They are, however, consistent with a position further north, close to either the east or west coasts of Gondwana; Vancouver Island could have been adjacent either to Arabia or to southern California, or to India and western Australia or Equador (Fig. 10, VN or VS). Alternatively it could, during the Late Paleozoic and Triassic, have been part of a block that was separate from Gondwana and which included Malaysia and possibly southern China. McElhinny *et al.* (1974) have given paleomagnetic evidence for supposing that Malaysia was separate from Gondwana at that time (M of Fig. 10).

Elements of the Late Paleozoic faunas of Vancouver Island, particularly some bryozoans and brachiopods, have affinities with the benthonic faunas of Timor and Australia (Fritz 1932; Crockford 1957; Yole 1963, 1965; Danner 1977). Triassic benthonic faunas of the western Cordillera, including Vancouver Island, indicate warm water conditions, presumably closer to the paleoequator than their counterparts of adjacent cratonic North America (Tozer 1970). Thus the paleomagnetic and biogeographical evidence are compatible with Vancouver Island in the Late Paleozoic and Triassic being situated in the southwest Pacific, either close to the eastern shore of

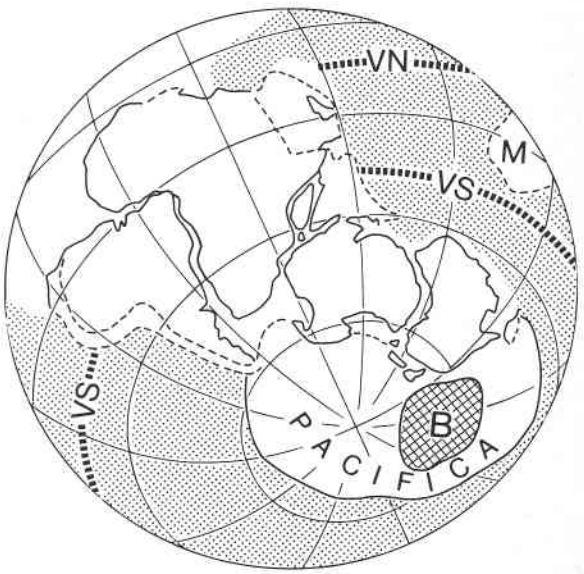


FIG. 10. Comparisons with Gondwana Triassic paleolatitudes are from Irving (1979*a*). VS and VN are the southern and northern paleolatitude options for Vancouver Island. B is the source of Wrangellia according to Nur and Ben-Avraham (1978). This map was actually drawn for Late Triassic (−200 Ma), but the Early Triassic (−225 Ma) map is not significantly different at the levels of accuracy discussed herein.

Gondwana or as part of a block situated to the east of Gondwana and that might have included Malaysia.

The four elements of Wrangellia (Wrangell Mountains, Vancouver Island, the Queen Charlotte Islands, and the Hells Canyon region of Oregon, Idaho, and Washington) are not in structural contact, and their former relative positions cannot be determined geologically. It is worthy of note, therefore, that the paleolatitudes of the Karmutsen ($18 \pm 6°$) and the Nikolai Greenstone (Hillhouse 1977) of the Wrangell Mountains ($15 \pm 5°$) are not significantly different. This implies that these two elements were closer together in the Late Triassic than at present, and that they have since been separated and smeared out along the fringe of the Cordillera. The earlier close proximity would also account for their remarkably similar stratigraphies (Monger 1977; Jones *et al.* 1977) and would mean that the Nikolai–Karmutsen volcanic event was of less areal extent than present distribution would imply. Alternatively, a wide distribution along the same paleolatitude zone is also possible but is inherently less likely.

The variations in declination of the *X* magnetization are greater than the variations in inclination (Fig. 4) and one possible explanation is rotation about local vertical axes (see section on *X* magnetizations). Elsewhere, large variations in declinations occur among different sampling areas in the Hazelton and Takla Groups and among the Topley intrusions of the Stikine terrane (Monger and Irving 1980). There is also a very large declination difference between the coeval Karmutsen and Nikolai Formations (Fig. 7). One of the features of the terranes of the western Cordillera is that in addition to the large displacements that they have undergone relative to North America, they may also have undergone differential rotations about vertical axes (Irving *et al.*, in press).

Acknowledgements

It is a pleasure to acknowledge G. A. Irving and A. Hicken for assistance in collecting samples; J. E. Hastie and P. Jones for help in making measurements, sample preparation, and data processing; J. W. H. Monger for many useful discussions; and J. L. Roy, D. B. Stone, and J. E. Muller for constructive criticism of the manuscript. The work was supported, in part, by grants to RWY by the National Research Council of Canada (A2137), the Department of Energy, Mines and Resources (45.65), and the Faculty of Graduate Studies and Research of Carleton University (125B) and these are gratefully acknowledged.

BURCHFIEL, B. C., and DAVIS, G. A. 1972. Structural framework and evolution of the southern part of the Cordilleran orogen, western United States. American Journal of Science, **272**, pp. 97–118.

CARLISLE, D. 1972. Late Paleozoic to mid-Triassic sedimentary–volcanic sequence of northwestern Vancouver Island. Report of Activities, Part B. Geological Survey of Canada, Paper 72-1B, pp. 24–30.

CARLISLE, D., and SUSUKI, T. 1974. Emergent basalt and submergent carbonate–clastic sequences including the Upper Triassic Dilleri and Welleri Zones on Vancouver Island. Canadian Journal of Earth Sciences, **11**, pp. 254–279.

CHURKIN, M. 1974. Paleozoic marginal ocean basin – volcanic arc systems in the Cordilleran fold belt. Society of Economic Paleontologists and Mineralogists, Special Publication 19, pp. 174–192.

CROCKFORD, J. 1957. Permian bryozoa from the Fitzroy Basin of Western Australia. Australian Bureau of Mineral Resources Bulletin, Geology and Geophysics, **34**. 134 p.

DANNER, W. R. 1977. Paleozoic rocks of NW Washington and adjacent parts of British Columbia. Society of Economic Paleontologists and Mineralogists, Pacific Section, Pacific Coast Paleogeography, Symposium 1, pp. 481–502.

DOUGLAS, R. J. W., GABRIELSE, H., WHEELER, J. O., STOTT, D. F., and BELYEA, H. R. 1970. Geology of western Canada. *In* Geology and economic minerals of Canada. *Edited by* R. J. W. Douglas. Geological Survey of Canada, Economic Geology Report No. 1, pp. 367–488.

FISHER, R. A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A, **217**, pp. 295–305.

FRITZ, M. A. 1932. Permian bryozoa from Vancouver Island. Royal Society of Canada Transactions, 3rd Series, **26**, pp. 93–109.

FYLES, J. T. 1955. Geology of the Cowichan Lake area, Vancouver Island. British Columbia Department of Mines, Bulletin 37. 72 p.

GABRIELSE, H., and WHEELER, J. O., *co-ordinators*. 1972. The Cordilleran Structural Province. *In* Variations in tectonic styles in Canada. *Edited by* R. A. Price and R. J. W. Douglas. Geological Association of Canada, Special Paper No. 11, pp. 1–81.

HICKEN, A., and IRVING, E. 1977. Tectonic rotations in western Canada. Nature, **268**, pp. 219–220.

HILDE, T. W. C., UYEDA, S., and KROENKE, K. 1976. Evolution of the western Pacific and its margins. *In* Geodynamics: progress and prospects. *Edited by* C. L. Drake. American Geophysical Union, pp. 1–15.

HILLHOUSE, J. W. 1977. Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy Quadrangle, Alaska. Canadian Journal of Earth Sciences, **14**, pp. 2578–2592.

IRVING, E. 1964. Paleomagnetism. J. Wiley and Sons, New York, NY. 399 p.

——— 1979*a*. Pole positions and continental drift since the Devonian. *In* The Earth—its origin and structure and evolution. *Edited by* M. W. McElhinny. Academic Press, New York, NY. pp. 567–593.

——— 1979*b*. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Canadian Journal of Earth Sciences, **16**, pp. 669–694.

IRVING, E., and PULLAIAH, G. 1976. Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of paleosecular variation in the Phanerozoic. Earth-Science Reviews, **12**, pp. 35–64.

IRVING, E., and YOLE, R. W. 1972. Paleomagnetism and the kinematic history of mafic and ultramafic rocks in fold mountain belts. Publications of the Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa, Ont., **42**, pp. 87–95.

IRVING, E., MONGER, W. H., and YOLE, R. W. In press. New paleomagnetic evidence for displaced terranes in British Columbia. *In* The continental crust and its mineral deposits. *Edited by* D. W. Strangway. Geological Association of Canada, Special Paper 20.

JELETZKY, J. A. 1976. Mesozoic and ? Tertiary rocks of Quatsino Sound, Vancouver Island, British Columbia, Geological Survey of Canada, Bulletin 242. 243 p.

JONES, D. L., IRWIN, W. P., and OVENSHINE, A. T. 1972. Southeastern Alaska—a displaced continental fragment? United States Geological Survey, Professional Paper 800-B, pp. B211–B217.

JONES, D. L., SILBERLING, N. J., and HILLHOUSE, J. 1977. Wrangellia—a displaced terrane in northwestern North America. Canadian Journal of Earth Sciences, **14**, pp. 2565–2577.

KUNIYOSHI, S. 1972. Petrology of the Karmutsen Group, northeastern Vancouver Island, British Columbia. Ph.D. thesis, University of California, Los Angeles, CA. 210 p.

KUNIYOSHI, S., and LIOU, J. G. 1976*a*. Contact metamorphism of the Karmutsen volcanics, Vancouver Island, British Columbia. Journal of Petrology, **17**, Part 1, pp. 73–99.

——— 1976*b*. Burial metamorphism of the Karmutsen volcanic rocks, northeastern British Columbia. American Journal of Science, **276**, pp. 1096–1119.

MCELHINNY, M. W. 1973. Palaeomagnetism and plate tectonics. Cambridge University Press, Cambridge, England. 358 p.

MCELHINNY, M. W., HAILE, N. S., and CRAWFORD, A. R. 1974. Paleomagnetic evidence shows Malay Peninsula was not part of Gondwanaland. Nature, **252**, pp. 641–645.

MONGER, J. W. H. 1977. Upper Paleozoic rocks of the western Canadian Cordillera and their bearing on Cordilleran evolution. Canadian Journal of Earth Sciences, **14**, pp. 1832–1859.

MONGER, J. W. H., and IRVING, E. 1980. Northward displacement of north-central British Columbia. Nature, **285**, pp. 289–293.

MONGER, J. W. H., and PRICE, R. A. 1979. Geodynamic evolution of the Canadian Cordillera—progress and problems. Canadian Journal of Earth Sciences, **16**, pp. 770–791.

MONGER, J. W. H., and ROSS, C. A. 1971. Distribution of Fusulinaceans in the western Canadian Cordillera. Canadian Journal of Earth Sciences, **8**, pp. 259–278.

MONGER, J. W. H., SOUTHER, J. G., and GABRIELSE, H. 1972. Evolution of the Canadian Cordillera: a plate-tectonic model. American Journal of Science, **272**, pp. 577–602.

MULLER, J. E. 1977*a*. Geology of Vancouver Island. Geological Association of Canada, 1977 Annual Meeting, Guidebook, Field Trip No. 7. 54 p.

——— 1977*b*. Evolution of the Pacific Margin, Vancouver Island, and adjacent regions. Canadian Journal of Earth Sciences, **14**, pp. 2062–2085.

MULLER, J. E., and CARSON, D. J. T. 1969. Geology and mineral deposits of Alberni map-area, British Columbia (92F). Geological Survey of Canada, Paper 68-50. 52 p.

MULLER, J. E., NORTHCOTE, K. E., and CARLISLE, D. 1974. Geology and mineral deposits of Alert Bay – Cape Scott map-area, Vancouver Island, British Columbia. Geological Survey of Canada, Paper 74-8. 77 p.

NUR, A., and BEN-AVRAHAM, Z. 1978. Speculations on mountain building and the lost Pacific continent. Journal of Physics of the Earth, **26**, pp. S21–S37.

PACKER, D. R., and STONE, D. B. 1974. Paleomagnetism of Jurassic rocks from southern Alaska, and the tectonic implications. Canadian Journal of Earth Sciences, **11**, pp. 976–997.

PATERSON, I. A. 1977. The geology and evolution of the Pinchi Fault zone at Pinchi Lake, central British Columbia. Canadian Journal of Earth Sciences, **14**, pp. 1324–1342.

ROY, J. L., SANDERS, E., and REYNOLDS, J. 1972. Un four électrique pour l'étude des propriétes magnetiques des roches. Publications of the Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa, Ont., **42**, pp. 229–237.

ROY, J. L., REYNOLDS, J., and SANDERS, E. 1973. An alternating field demagnetizer for rock magnetism studies. Publications of the Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa, Ont., **44**, pp. 37–45.

SCHWARZ, E. J., MULLER, J. E., and CLARK, K. R. 1980. Paleomagnetism of the Karmutsen basalts from southeast Vancouver Island. Canadian Journal of Earth Sciences, **17**, pp. 389–399.

SOUTHER, J. G. 1977. Volcanism and tectonic environments in the Canadian Cordillera—a second look. *In* Volcanic regimes in Canada. *Edited by* W. R. A. Baragar, L. C. Coleman, and J. M. Hall. Geological Association of Canada, Special Paper 16, pp. 3–24.

STONE, D. B. 1979. Paleomagnetism and paleogeographic reconstruction of southern Alaska. *In* Proceedings of the 6th annual Geological Society of Alaska Symposium, Anchorage, Alaska. *Edited by* A. Sisson. pp. K1–K17.

STONE, D. B., and PACKER, D. R. 1977. Tectonic implications of Alaskan paleomagnetic data. Tectonophysics, **37**, pp. 183–201.

——— 1979. Paleomagnetic data from the Alaska Peninsula. Geological Society of America Bulletin, **90**, pp. 545–560.

SUTHERLAND-BROWN, A. 1966. Tectonic history of the Insular Belt of British Columbia. *In* Tectonic history and mineral deposits of the western Cordillera. *Edited by* H. C. Gunning. Canadian Institute of Mining and Metallurgy, Special Volume 8, pp. 83–100.

SYMONS, D. T. A. 1971*a*. Paleomagnetism of the Jurassic Island Intrusions, British Columbia. Geological Survey of Canada, Paper 70-63, pp. 1–17.

——— 1971*b*. Paleomagnetic notes on the Karmutsen basalts, Vancouver Island, British Columbia. Geological Survey of Canada, Paper 71-24, pp. 9–24.

TOZER, E. T. 1970. Marine Triassic faunas. *In* Geology and economic minerals of Canada. *Edited by* R. J. W. Douglas. Geological Survey of Canada, Economic Geology Report No. 1, pp. 633–640.

YOLE, R. W. 1963. An early Permian fauna from Vancouver Island, British Columbia. Bulletin of Canadian Petroleum Geology, **11**, pp. 138–149.

——— 1965. A faunal and stratigraphic study of Upper Paleozoic rocks of Vancouver Island, B.C. Ph.D. thesis, University of British Columbia, Vancouver, B.C. 254 p.

——— 1969. Upper Paleozoic stratigraphy of Vancouver Island, British Columbia. Proceedings of the Geological Association of Canada, **20**, pp. 30–40.

ZIJDERVELD, J. D. A. 1975. Paleomagnetism of the Esterel rocks. Doctoral thesis, University of Utrecht. 197 p.

Appendix: Tectonic Analysis of Paleomagnetic Results from Orogenic Belts

A direction of magnetization observed at a sampling site can be represented by a virtual geomagnetic pole (VGP). The method for calculating VGP's may be found in Irving (1964, p. 43) or McElhinny (1973, p. 25). Directions observed at several sampling sites in a rock unit can be averaged to obtain the mean direction with declination and inclination (D, I) and circle of confidence radius α (Fisher 1953). By convention, the north-seeking

directions are always specified. The north-seeking directions point to the geographic north (south) in a field of normal (reversed) polarity. The directions at each site are restored to original directions by correcting for geologically observed attitudes if these are not horizontal. The latitude and longitude of the paleopole (λ, ϕ) for the rock unit and its error can be obtained either directly from D, I, and α of the rock unit (model A paleopole), or by averaging the VGP's for each site using Fisher's statistics to calculate the radius of the polar error A (model B paleopole; refer to Irving (1964, p. 49) for description of the A and B models of the geomagnetic field from which these procedures are derived). Differences between results obtained by the two methods are small enough to be neglected in studies of the large effects currently being observed in orogenic belts. For simplicity of calculation and argument, we deal only with model B calculations.

The mean declination (D) is measured clockwise from present geographic north. D is a measure of the sum of all the rotations (ψ) that the test rock unit has undergone relative to the direction of present geographic north since the time the magnetization was acquired. I is measured relative to restored paleohorizontal and can be used to calculate the paleolatitude λ_p of the rock unit at the time it was formed; ψ and λ_p can be accurately estimated only if the geological attitudes are known. Because of reversals of the field, the sign of paleolatitude determined from the observations themselves is ambiguous; without precise dating and a complete reversal chronology, it is not possible to determine whether the rock unit (latitude Λ, longitude Φ) was originally formed in the southern or northern hemispheres. Finally, paleolongitude cannot be determined directly by paleomagnetism.

Paleopoles obtained from several rock units distributed over the surface of a rigid craton or plate and formed during a given interval of time can be combined to obtain a reference paleopole (λ_r, ϕ_r) with associated error circle of radius A_r. Then λ_r, ϕ_r are the coordinates of the paleopole that would be expected from a study of other rock units formed during the same time interval on the same rigid craton. Should the test locality not be on the craton but in an adjacent orogenic belt, it may have moved relative to the craton, so the paleopole λ, ϕ observed from it will not agree with the expected paleopole λ_r, ϕ_r. The significance of the disagreement can, in an approximate fashion, be studied by constructing standard error circles about the reference and test paleopoles. If the circles do not intersect (do intersect) the two are (are not) likely to be significantly different and the test locality has probably moved (probably has not moved) relative to the craton. Note that motions in a paleolongitudinal sense may have occurred even if the test and reference poles are found to agree.

When a significant difference between the expected and test paleopoles is observed, it becomes necessary to express the difference in terms useful for tectonic analysis. This can be done by resolving the difference into two components of motion, referred to as RPD and RR. A displacement of the test paleopole away from or towards the expected reference paleopole may be caused by motions relative to the craton in a paleolatitudinal sense (relative paleolatitudinal displacements, RPD, of magnitude $\Delta°$). Displacement of the test paleopole to the right or left of the reference paleopole may be caused by rotation of the test locality about local vertical axes relative to the craton (relative rotation, RR, of magnitude $\Psi°$). Displacement in a paleolongitudinal sense yields no displacement of the paleopole.

Relative Paleolatitudinal Displacement (RPD)

The direction D, I and corresponding paleopole AP (λ, ϕ) are obtained from a test rock unit at test locality T (Λ, Φ) situated in an orogenic zone adjacent to a stable continental craton (Fig. A1). The observed paleolatitude of T is given by

$$[A1] \quad \tan\lambda_p = \pm\tfrac{1}{2}\tan I$$

and the uncertainty in paleolatitude ($d\lambda_p$) is equal to the error A in the paleopole

$$[A2] \quad d\lambda_p = A$$

The expected paleolatitude $\lambda_{p(exp)}$ of T calculated from the reference paleopole R(λ_r, ϕ_r) for the stable continental craton is given by

$$[A3] \quad \sin\lambda_{p(exp)} = \sin\Lambda\sin\lambda_r + \cos\Lambda\cos\lambda_r\cos(\phi_r - \phi)$$

For post mid-Carboniferous times the polarity of R is known because the continuity of the APW path

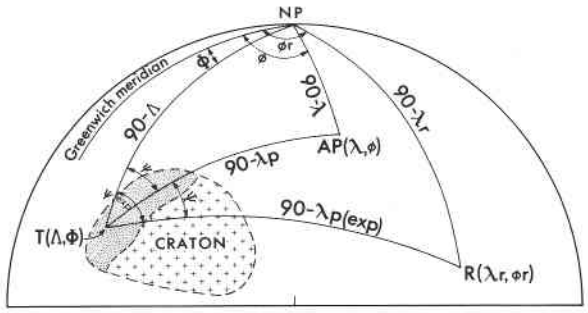


FIG. A1. Observed and expected paleolatitudes λ_p and $\lambda_{p(exp)}$.

with the present geographic north pole has been established for the major cratons. The polarity of R becomes progressively uncertain further backward in time. The error in [3] is equal to the error A_r in the reference pole so that

[A4] $d\lambda_{p(exp)} = A_r$

Hence, the mean relative paleolatitudinal displacement, RPD, of magnitude Δ (Fig. A2) is given by

[A5] $\Delta = \lambda_{p(exp)} - \lambda_p$

and an error $d\Delta$ then can be obtained from

[A6] $d\Delta = (A^2 + A_r^2)^{1/2}$

Paleolatitudes are positive (negative) when north (south), and positive (negative) values of [A5] indicate that displacements are towards the north (south). Equation [A5] will always have two solutions because of the ambiguity in the sign of λ_p.

Relative Rotation (RR)

The declination D is a measure of the sum of the rotations ψ of the test locality relative to the direction of present geographic north

[A7] $\psi = D$ or $D + 180$

the ambiguity arising because of reversals of the field. The direction may point to the geographic north or south pole. The error $d\psi$ (Fig. A3) is given by

[A8] $\sin d\psi = \sin A/(1 - \cos^2 A \sin^2 \lambda_p)^{1/2}$

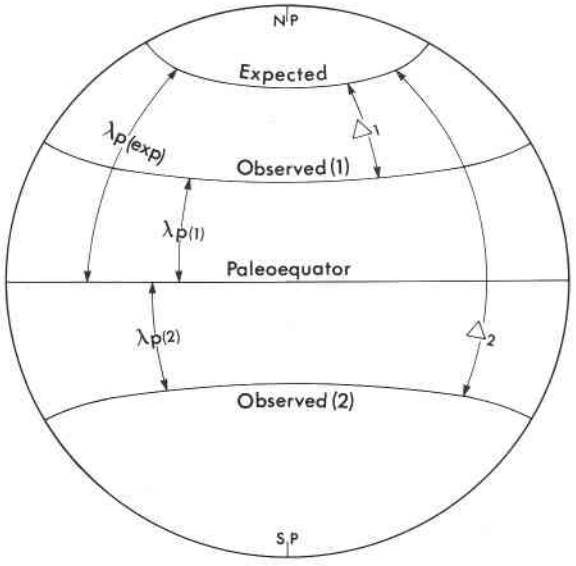


FIG. A2. Relative paleolatitude displacement (RPD). The observed and expected paleolatitudes and the two possible values Δ_1 and Δ_2 for the RPD are shown.

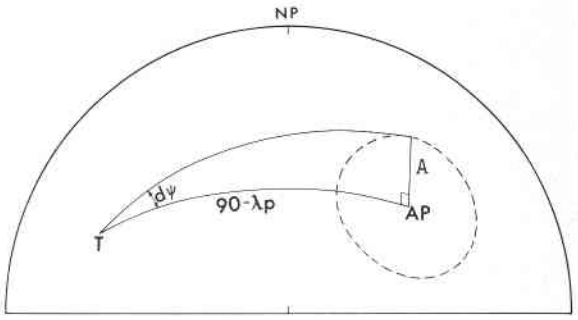


FIG. A3. Errors in the observed and expected rotations ($d\psi$).

The expected rotation ψ_{exp} of the test rock unit relative to present geographic north (Fig. A1), assuming it has remained unmoved relative to the craton, is given by

[A9] $\sin \psi_{exp} = (\sin(\phi_r - \phi) \cos \lambda_r)/\cos \lambda_{p(exp)}$

and the error $d\psi_{exp}$ by

[A10] $\sin d\psi_{exp} = \sin A_r/(1 - \cos^2 A_r \sin^2 \lambda_{p(exp)})$

Hence Ψ, the relative rotation, is obtained from

[A11] $\Psi = \psi_{exp} - \psi$

and the error $d\Psi$ is given by

[A12] $d\Psi = (d\psi^2 + d\psi^2_{exp})^{1/2}$

ψ and ψ_{exp} are measured clockwise relative to present geographic north and a negative (positive) value of Ψ indicates a clockwise (anticlockwise) rotation. There is always an uncertainty of 4π in the RR values, which may be equal to Ψ, $\Psi + 4\pi$, etc., although extreme values are geologically implausible. It should also be noted that a single result from one test rock unit does not allow a distinction to be made between a rotation of Ψ clockwise and one of $4\pi - \Psi$ anticlockwise.

Cautionary Note

RPD and RR are geometrical concepts and may not be directly relatable to actual tectonic motions. If RPD's are negligible and only RR's are significant, we can be reasonably sure that only rotations about local vertical axes have occurred. If RR's are negligible and significant, RPD's are present, and we can be reasonably sure that displacements in a paleolatitudinal sense have occurred without rotations. However, if both RR's and RPD's are present and significant, they may or may not arise from separate tectonic causes. For example, the oblique convergence of an exotic terrane towards the western Cordillera of North America from the southwest could produce a paleomagnetic signature (far-sided, right-handed paleopoles)

identical to that of a northward RPD coupled with a clockwise RR.

Example

The following values have been observed for the X magnetization of the Karmutsen Formation: $\Lambda, \Phi = 50.0°N, 125.5°W$ (Fig. 1); $D, I = 008, -33$ (Table 2, entry 13); $\lambda, \phi, A = 21°N, 044°E, 6°$ (Table 2, entry 14); and $\lambda_r, \phi_r, A_r = 68°N, 093°E, 6°$ (Irving 1979*b*). From [A1] the paleolatitude λ_p of the Karmutsen Formation when it was laid down was either 18°S, if the field were normal, or 18°N, if reversed. The error in the paleolatitude from [A2] is 6° ($P = 0.05$). The paleolatitude that would be observed if Vancouver Island had remained fixed relative to North America by [A3] and [A4] is 31 ± 6°N. Hence by [A5] and [A6], the RPD of Vancouver Island relative to cratonic North America is either 49 ± 8° or 13 ± 8° to the north. The observed sum of the rotations of the Karmutsen Formation locality relative to the direction of the present geographic north is, by [A7] and [A8], equal to 8 ± 6°, if the field were normal, and 188 ± 6°, if reversed. The rotation that would be observed if Vancouver Island had remained fixed relative to North America, by [A9] and [A10], is either 14 ± 7° anticlockwise, if normal, or 166 ± 7° clockwise, if reversed. Hence by [A11] and [A12], the RR of Vancouver Island relative to cratonic North America is 22 ± 9° clockwise, if the field were normal, or 202 ± 9° clockwise, if reversed. For the reasons explained above we believe that a normal field configuration yields the most probable solution so that $\Delta = 49 \pm 8°$ and $\Psi = 22 \pm 9°$ clockwise. Since we do not yet have a time sequence of paleomagnetic results from Vancouver Island, the arguments are inconclusive, and the alternative solution given by the reversed field configuration of $\Delta = 13 \pm 8°$ and $\Psi = 158 \pm 9°$ anticlockwise is also possible.